IV B.TECH – I SEMESTER

Software Architecture And Design Patterns

Professional Elective-IV | JNTUA R23 · B.Tech CSE

	SOFTWARE ARCHITECTURE AND DESIGN PATTERNS (Professional Elective –IV)	L	T	P	C
		3	0	0	3

Course Objectives:

After completing this course, the student should be able to:

- To under stand the concept of patterns and the Catalog.
- To discuss the Presentation tier design patterns and their affect on: sessions, client access, validation and consistency.
- To understand the variety of implemented bad practices related to the Business and Integration tiers.

Course Out comes:

- To highlight the evolution of patterns.
- To learn how to add functionality to designs while minimizing complexity
- To learn what design patterns really are, and are not
- To know about specific design patterns.
- To learn how to use design patterns to keep code quality high without over design.

UNIT-I

Envisioning Architecture: The Architecture Business Cycle, What is Software Architecture, Architectural patterns, reference models, reference architectures, architectural structures and views. Creating an Architecture: Quality Attributes, Achieving qualities, Architectural styles and patterns, designing the Architecture, Documenting software architectures, Reconstructing Software Architecture.

UNIT- II

Analyzing Architectures: Architecture Evaluation, Architecture design decision making, ATAM, CBAM. Moving from one system to many: Software Product Lines, Building systems from off the shelf components, Software architecture in future.

UNIT- III

Patterns: Pattern Description, Organizing catalogs, role in solving design problems, Selection and usage.Creational and Structural patterns: Abstract factory, builder, factory method, prototype, singleton, adapter, bridge, composite, façade, flyweight.

UNIT- IV

Behavioural patterns: Chain of responsibility, command, Interpreter, iterator, mediator, memento, observer, state, strategy. template method, visitor.

UNIT- V

Case Studies: A-7E – A case study in utilizing architectural structures, The World Wide Web - a case study in interoperability, Air Traffic Control – a case study in designing for high availability, Celsius Tech – a case study in product line development.

TEXT BOOKS:

- Software Architecture in Practice, second edition, Len Bass, Paul Clements & Rick Kazman, Pearson Education, 2003.
- Design Patterns, Erich Gamma, Pearson Education.

REFERENCES:

- Beyond Software architecture, Luke Hohmann, Addison wesley, 2003.
- Software architecture, David M. Dikel, David Kane and James R. Wilson, Prentice Hall PTR, 2001
- Software Design, David Budgen, second edition, Pearson education, 2003
- Head First Design patterns, Eric Freeman & Elisabeth Freeman, O'REILLY, 2007.
- Design Patterns in Java, Steven John Metsker & William C. Wake, Pearson education, 2006 6. J2EE Patterns, Deepak Alur, John Crupi & Dan Malks, Pearson education, 2003.
- Design Patterns in C#, Steven John metsker, Pearson education, 2004.
- Pattern Oriented Software Architecture, F.Buschmann &others, John Wiley & Sons.